「Galois group of $(x^2-2)(x^2-3)$ over $\mathbb{Q}$.

Since the polynomial $(x^2-2)(x^2-3)$ has four distinct roots $\pm\sqrt{2}, \pm\sqrt{3}$, it is separable.
And, since $\sqrt{2} \notin \mathbb{Q}(\sqrt{3})$, $[\mathbb{Q}(\sqrt{2},\sqrt{3}):\mathbb{Q}] = [\mathbb{Q}(\sqrt{2},\sqrt{3}), \mathbb{Q}(\sqrt{3})]\cdot[\mathbb{Q}(\sqrt{3}):\mathbb{Q}] = 4$.
So, $|\mathrm{Aut}(K/\mathbb{Q})| = [K:\mathbb{Q}] = 4$ where $K = \mathbb{Q}(\sqrt{2},\sqrt{3})$, the splitting field over $\mathbb{Q}$.
Since the automorphism leaving $\mathbb{Q}$ fixed must be conjugate the roots,
there are four possible automorphisms;

$$\begin{cases}\sqrt{2}\mapsto\sqrt{2}\\ \sqrt{3}\mapsto\sqrt{3}\end{cases},\quad \begin{cases}\sqrt{2}\mapsto-\sqrt{2}\\ \sqrt{3}\mapsto\sqrt{3}\end{cases},\quad \begin{cases}\sqrt{2}\mapsto\sqrt{2}\\ \sqrt{3}\mapsto-\sqrt{3}\end{cases},\quad \begin{cases}\sqrt{2}\mapsto-\sqrt{2}\\ \sqrt{3}\mapsto-\sqrt{3}.\end{cases}$$

Let $\sigma : K \to K ; a+b\sqrt{2}+c\sqrt{3}+d\sqrt{6} \mapsto a-b\sqrt{2}+c\sqrt{3}-d\sqrt{6}$

$\tau : K \to K ; a+b\sqrt{2}+c\sqrt{3}+d\sqrt{6} \mapsto a+b\sqrt{2}-c\sqrt{3}-d\sqrt{6}$.

(That is, $\sigma : \begin{cases}\sqrt{2}\mapsto-\sqrt{2}\\ \sqrt{3}\mapsto\sqrt{3}\end{cases}$ and $\tau : \begin{cases}\sqrt{2}\mapsto\sqrt{2}\\ \sqrt{3}\mapsto-\sqrt{3}.\end{cases}$, and we can observe that

$\sigma\tau : \begin{cases}\sqrt{2}\mapsto-\sqrt{2}\\ \sqrt{3}\mapsto-\sqrt{3}.\end{cases}$ $\tau\sigma : \begin{cases}\sqrt{2}\mapsto-\sqrt{2}\\ \sqrt{3}\mapsto-\sqrt{3}\end{cases}$, so $\sigma^2 = \tau^2 = 1$ and $\sigma\tau = \tau\sigma$

Now, $\mathrm{Gal}(K/\mathbb{Q}) = \{1, \sigma, \tau, \sigma\tau\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, corresponding σ to $(1,0)$ and τ to $(0,1)$.
Moreover,

$$\begin{tikzcd}
 & \mathbb{Q}(\sqrt{2},\sqrt{3}) = K^{\{id\}} \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] & \\
\mathbb{Q}(\sqrt{2}) = K^{\langle\tau\rangle} \arrow[dr, dash] & \mathbb{Q}(\sqrt{6}) = K^{\langle\sigma\tau\rangle} \arrow[d, dash] & \mathbb{Q}(\sqrt{3}) = K^{\langle\sigma\rangle} \arrow[dl, dash] \\
 & \mathbb{Q} = K^{\mathrm{Gal}(K/\mathbb{Q})} &
\end{tikzcd}$$

Note: $\sqrt{2} \in \mathbb{Q}(\sqrt{3})$ because: if $\sqrt{2} = a + b\sqrt{3}$ for some $a, b \in \mathbb{Q}$, then $2 = a^2 + 3b^2 + 2ab\sqrt{3}$.
it is contradiction

「Galois group of x^3-2 over $\mathbb{Q}$.

Denote $\zeta_3 = e^{i\cdot\frac{2\pi}{3}} = \frac{1}{2}(-1+i\cdot\sqrt{3})$, the 3th primitive root of unity.

Since $\zeta_3^3=1$, the set $\{\sqrt[3]{2}, \sqrt[3]{2}\,\zeta_3, \sqrt[3]{2}\cdot\zeta_3^2\}$ are all roots of x^3-2 over $\mathbb{Q}$.
So, the splitting field is $K=\mathbb{Q}(\sqrt[3]{2},\zeta_3). = \mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})$.
observe that: the minimal polynomial of $\sqrt[3]{2}$ over $\mathbb{Q}$ is x^3-2, and
" ζ_3 over $\mathbb{Q}$ is x^2+x+1.
And, since x^2+x+1 is irreducible over $\mathbb{Q}(\sqrt[3]{2})$, (it cannot have root in $\mathbb{Q}(\sqrt[3]{2})$)

$$[\mathbb{Q}(\sqrt[3]{2},\zeta_3):\mathbb{Q}] = [\mathbb{Q}(\sqrt[3]{2},\zeta_3):\mathbb{Q}(\sqrt[3]{2})]\cdot[\mathbb{Q}(\sqrt[3]{2}):\mathbb{Q}] = 2\cdot 3 = 6.$$

the tower Lemma gave. And, since x^3-2 is separable, $|\mathrm{Aut}(K/\mathbb{Q})| = [K:\mathbb{Q}] = 6$.
Now, Construct explicity the $\mathrm{Aut}(K/\mathbb{Q})$.
Since any automorphism of K leaving $\mathbb{Q}$ fixed must Conjugate the roots,
it must maps $\sqrt[3]{2}$ to $\sqrt[3]{2}$ or $\sqrt[3]{2}\,\zeta_3$ or $\sqrt[3]{2}\cdot\zeta_3^2$
and maps ζ_3 to ζ_3 or ζ_3^2.
So, there are exactly 6 possiblities. Consider two automorphisms

σ satisfying $\sigma(\sqrt[3]{2}) = \sqrt[3]{2}\cdot\zeta_3$ and $\sigma(\zeta_3)=\zeta_3$

and τ satisfying $\tau(\sqrt[3]{2}) = \sqrt[3]{2}$ and $\tau(\zeta_3) = \zeta_3^2 = -1-\zeta_3$.

Then, $\sigma^3: \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2}\,\zeta_3 \mapsto \sqrt[3]{2}\cdot\zeta_3^2 \mapsto \sqrt[3]{2}\,\zeta_3^3 = \sqrt[3]{2} \\ \zeta_3 \mapsto \zeta_3. \end{cases}$, $\tau^2: \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2} \\ \zeta_3 \mapsto \zeta_3^2 \mapsto \zeta_3^4 = 1-\zeta_3 = \zeta_3 \end{cases}$

are actually identities. Moreover,

$\sigma^2: \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2}\cdot\zeta_3^2 \\ \zeta_3 \mapsto \zeta_3 \end{cases}$, $\sigma\tau: \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2}\cdot\zeta_3 \\ \zeta_3 \mapsto \zeta_3^2 \end{cases}$, $\sigma^2\tau: \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2}\,\zeta_3^2 \\ \zeta_3 \mapsto \zeta_3^2 \end{cases}$, $\tau\sigma: \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2}\cdot\zeta_3^2 \\ \zeta_3 \mapsto \zeta_3^2 \end{cases}$

So, $\mathrm{Aut}(K/\mathbb{Q}) = \{1, \sigma, \sigma^2, \tau, \sigma\tau, \sigma^2\tau\} \cong S_3$, Corresponding σ to $(1\,2\,3)$, τ to $(1\,2)$
Moreover,

$$\begin{tikzcd}[column sep=small]
 & & \mathbb{Q}(\sqrt[3]{2},\zeta_3) = K^{\{id\}} \arrow[dll, dash] \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] & \\
\mathbb{Q}(\zeta_3) = K^{\langle\sigma\rangle} \arrow[drr, dash] & \mathbb{Q}(\sqrt[3]{2}) = K^{\langle\tau\rangle} \arrow[dr, dash] & \mathbb{Q}(\sqrt[3]{2}\cdot\zeta_3^2) = K^{\langle\sigma\tau\rangle} \arrow[d, dash] & \mathbb{Q}(\sqrt[3]{2}\cdot\zeta_3) = K^{\langle\sigma^2\tau\rangle} \arrow[dl, dash] \\
 & & \mathbb{Q} = K^{\langle\sigma,\tau\rangle} &
\end{tikzcd}$$

How to find a fixed field corresponding to $\langle\sigma\tau\rangle$?
As we know, the $\mathrm{Tr}_{K/K^{\langle\sigma\tau\rangle}}$ is invariant under any automorphism in $\langle\sigma\tau\rangle$.
So, Calculate:

$$\mathrm{Tr}(\sqrt[3]{2}) = id(\sqrt[3]{2}) + (\sigma\circ\tau)(\sqrt[3]{2}) = \sqrt[3]{2} + \sqrt[3]{2}\,\zeta_3 = \sqrt[3]{2}(1+\zeta_3) = \sqrt[3]{2}\cdot(-\zeta_3^2).$$

Therefore, $\sqrt[3]{2}\,\zeta_3^2$ is fixed by id and $\sigma\circ\tau$, so $\mathbb{Q}(\sqrt[3]{2}\cdot\zeta_3^2) \subseteq K^{\langle\sigma\tau\rangle}$.
Moreover, $[\mathbb{Q}(\sqrt[3]{2},\zeta_3):\mathbb{Q}(\sqrt[3]{2}\cdot\zeta_3^2)] = 2$, so the Galois theorem gives the result.

「Galois group of x^8-2 over $\mathbb{Q}$.

By the eisenstein criterion, $2\in 2\mathbb{Z}$ but $2\notin 2^2\mathbb{Z}$, so x^8-2 is irreducible over $\mathbb{Q}$.
And, the formal derivative $8\cdot x^7$ has only root as zero, but $0^8-2\neq 0$,
So the given polynomial is separable. The polynomial x^8-2 has 8 distinct roots,

$$\sqrt[8]{2}\cdot\zeta_8^k \quad (k=0,1,\cdots,7) \quad \text{where } \zeta_8=\exp\left(2\pi i\cdot\frac{1}{8}\right)=\frac{\sqrt{2}}{2}(1+i)$$

Therefore, the splitting field $K=\mathbb{Q}(\sqrt[8]{2},\cdots,\sqrt[8]{2}\,\zeta_8^7)=\mathbb{Q}(\sqrt[8]{2},i)$, because
the $\subseteq$ inclusion is given by $(\sqrt[8]{2})^4=\sqrt{2}$, so $\zeta_8\in\mathbb{Q}(\sqrt[8]{2},i)$, and
the $\supseteq$ inclusion is given by $\zeta_8^2=\frac{1}{2}(2i)=i\in\mathbb{Q}(\sqrt[8]{2},\cdots,\sqrt[8]{2}\cdot\zeta_8^7)$.
By the irreducibility, the minimal polynomial of $\sqrt[8]{2}$ is x^8-2, and
the minimal polynomial of i is x^2+1 trivially. Since $\mathbb{Q}(\sqrt[8]{2})\subseteq\mathbb{R}$, $i\notin\mathbb{Q}(\sqrt[8]{2})$,
thus the tower lemma gives

$$[K:\mathbb{Q}]=[\mathbb{Q}(\sqrt[8]{2},i):\mathbb{Q}(\sqrt[8]{2})]\cdot[\mathbb{Q}(\sqrt[8]{2}):\mathbb{Q}]=16.$$

Find explicitly $Gal(K/\mathbb{Q})$. Since an automorphism in $Gal(K/\mathbb{Q})$ conjugates the $\sqrt[8]{2}$ and i,
therefore, we can define

$$\sigma:\begin{cases}\sqrt[8]{2}\mapsto\sqrt[8]{2}\cdot\zeta_8\\ i\mapsto i\end{cases} \quad\text{and}\quad \tau:\begin{cases}\sqrt[8]{2}\mapsto\sqrt[8]{2} & \left(\sigma(\zeta_8)=\frac{1+i}{2}\cdot\sigma((\sqrt[8]{2})^4)=\zeta_8^5\right)\\ i\mapsto -i & \left(\tau(\zeta_8)=\tau\left(\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}i\right)=\frac{\sqrt{2}}{2}-\frac{\sqrt{2}}{2}i=\zeta_8^7\right)\end{cases}$$

Then, just simple calculation gives $id,\sigma,\cdots,\sigma^7,\tau,\tau\sigma,\cdots,\tau\sigma^7$ are mutually distinct. So, $Gal(K/\mathbb{Q})=\{id,\sigma,\cdots,\sigma^7,\tau,\cdots,\tau\sigma^7\}$. Further, observe that

$$\sigma^8=\tau^2=1 \quad\text{and}\quad \sigma\cdot\tau:\begin{cases}\sqrt[8]{2}\mapsto\sqrt[8]{2}\cdot\zeta_8 & \text{and}\\ i\mapsto -i\end{cases}$$

$$\tau\sigma^3:\begin{cases}\overset{0}{\sqrt[8]{2}}\overset{\sigma}{\mapsto}\overset{1}{\sqrt[8]{2}\cdot\zeta_8}\overset{\sigma}{\mapsto}\overset{1+5}{\sqrt[8]{2}\cdot\zeta_8^6}\overset{\sigma}{\mapsto}\overset{1+30}{\sqrt[8]{2}\cdot\zeta_8^7}\overset{\tau}{\mapsto}\sqrt[8]{2}\cdot\zeta_8\\ i\mapsto -i\end{cases}$$

Therefore $Gal(K/\mathbb{Q})=\langle\sigma,\tau\mid\sigma^8=\tau^2=1,\ \sigma\cdot\tau=\tau\sigma^3\rangle$, the quasihedral group.

$$\begin{array}{l} Gal(K/\mathbb{Q})\\ \ \ 2\,|\\ \langle\sigma\rangle\\ \ \ 2\,|\\ \langle\sigma^2\rangle\\ \ \ 2\,|\\ \langle\sigma^4\rangle\\ \ \ 2\,|\\ \{id\}\end{array}$$

「Galois group of x^5-3 over $\mathbb{Q}(\zeta)$, where ζ is a primitive fifth root of unity.

Observe that: x^5-3 has five distinct roots, $\sqrt[5]{3}\cdot\zeta^k$ $(k=0,1,\cdots,4)$.
Since the minimal polynomial of ζ over $\mathbb{Q}$ is $x^4+x^3+x^2+x+1$,
therefore $[\mathbb{Q}(\zeta):\mathbb{Q}]=4$. And, the minimal polynomial of $\sqrt[5]{3}$ over $\mathbb{Q}$ is x^5-3,
therefore $[\mathbb{Q}(\sqrt[5]{3}):\mathbb{Q}]=5$. Therefore, the splitting field of x^5-3 over $\mathbb{Q}$,

$$K=\mathbb{Q}(\sqrt[5]{3},\ \sqrt[5]{3}\zeta^1,\ \sqrt[5]{3}\zeta^2,\ \sqrt[5]{3}\zeta^3,\ \sqrt[5]{3}\zeta^4)=\mathbb{Q}(\sqrt[5]{3},\zeta)=\mathbb{Q}(\sqrt[5]{3})\mathbb{Q}(\zeta)$$

has degree of 20 over $\mathbb{Q}$, that is, $[K:\mathbb{Q}]=20$.
More precisely, by the tower lemma, $[K:\mathbb{Q}]$ is divided by $[\mathbb{Q}(\sqrt[5]{3}):\mathbb{Q}]$ and $[\mathbb{Q}(\zeta):\mathbb{Q}]$.
Since 5 and 4 are relatively prime, $[K:\mathbb{Q}]\geq 20$.
And, $x^5-3\in\mathbb{Q}(\zeta)[x]$ has $\sqrt[5]{3}$ as a root, so

$$[\mathbb{Q}(\sqrt[5]{3},\zeta):\mathbb{Q}(\zeta)]\cdot[\mathbb{Q}(\zeta):\mathbb{Q}]\leq 20.$$

Consequently, $[K:\mathbb{Q}]=20$. Now, by the Fundamental Theorem of Galois Theory,
$K/\mathbb{Q}(\zeta)$ is also Galois, with $[K:\mathbb{Q}(\zeta)]=5$.
Therefore, $\mathrm{Gal}(K/\mathbb{Q}(\zeta))$ has order of 5, so it must be a cyclic group of order 5. Consequently, the automorphism

$$\sigma:\begin{cases}\sqrt[5]{3}\mapsto\sqrt[5]{3}\cdot\zeta\\ \zeta\mapsto\zeta\end{cases}$$ (It is given by the Conjugate isomorphism theorem).

has order of 5, therefore $\mathrm{Gal}(K/\mathbb{Q}(\zeta))=\langle\sigma\rangle\cong\mathbb{Z}_5$. ☑

Lattiscally,

$$\begin{tikzcd}
 & \mathbb{Q}(\sqrt[5]{3},\zeta)=K^{\{id\}} \arrow[dl, dash, "5"'] \arrow[dr, dash, "4"] & \\
\mathbb{Q}(\zeta)=K^{\langle\sigma\rangle} \arrow[dr, dash, "4"'] & & \mathbb{Q}(\sqrt[5]{3})=K^{\langle\tau\rangle} \arrow[dl, dash, "5"] \\
 & \mathbb{Q}=K^{\mathrm{Aut}(K/\mathbb{Q})=\langle\sigma,\tau\rangle} &
\end{tikzcd}$$

where $\tau:\begin{cases}\sqrt[5]{3}\mapsto\sqrt[5]{3}\\ \zeta\mapsto\zeta^2.\end{cases}$ Therefore, $\mathrm{Aut}(K/\mathbb{Q})\cong\langle(1\ 2\ 3\ 4\ 5),(2\ 3\ 5\ 4)\rangle$.

[Galois group of x^6+3 over $F=\mathbb{Q}$ or $\mathbb{F}_5$ or $\mathbb{F}_7$. (UCLA Qual, 2020 Fall)
For each case, determine the splitting field K and evaluate $[K:F]$.
And, find the Galois group $Gal(K/F)$.
(Note: the field of characteristic 0 or finite field are perfect.).

1) $F=\mathbb{Q}$.
By the eisenstein creterion, all coefficient of x^6+3 except the leading coefficients are contained in $3\mathbb{Z}$, but the square of constant term $3\notin 3^2\mathbb{Z}$, the given polynomial x^6+3 is irreducible over $\mathbb{Q}$. The x^6+3 has 6 roots,

$$\sqrt[6]{3}\cdot w^{2k+1} \ (k=0,1,\cdots,5) \text{ where } w=\exp\left(\frac{i\pi}{6}\right)=\frac{\sqrt{3}}{2}+i\cdot\frac{1}{2}.$$

Therefore, the splitting field is $K=\mathbb{Q}(\sqrt[6]{3}\cdot w, \sqrt[6]{3}\cdot w^3, \cdots, \sqrt[6]{3}\cdot w^{11})$,
Clearly $\mathbb{Q}(\sqrt[6]{3}\cdot w)\subseteq K$, by the definition of generated field.
Meanwhile, $(\sqrt[6]{3}\cdot w)^3=\sqrt{3}\cdot w^3=\sqrt{3}\cdot\exp(i\pi/2)=\sqrt{3}\cdot i$ and $w^2=\frac{1}{2}+\frac{i\sqrt{3}}{2}$ implies $w^2\in\mathbb{Q}(\sqrt[6]{3}\cdot w)$, so $K=\mathbb{Q}(\sqrt[6]{3}\cdot w)$. Finally, since x^6+3 is irreducible, $[K:\mathbb{Q}]=6$.
Moreover, $Gal(K/\mathbb{Q})\cong S_3$, because: define two automorphisms on K as:
let $\theta=\sqrt[6]{3}\cdot w$ and $\lambda=w^2$. Then, $\theta^3=i\sqrt{3}$ and $\lambda=\frac{1}{2}+\frac{\theta^3}{2}$. Now, let

$$\sigma:\theta\mapsto\theta\cdot\lambda \quad\text{and}\quad \tau:\theta\mapsto\theta\lambda^2.$$

Then, $\sigma(\lambda)=\sigma(\frac{1}{2}+\frac{1}{2}\theta^3)=\frac{1}{2}+\frac{1}{2}\cdot\theta^3\cdot\lambda^3=\frac{1}{2}+\frac{1}{2}\cdot i\sqrt{3}\cdot w^6=\frac{1}{2}(1-i\sqrt{3})=w^{10}=\lambda^{-1}$.
So $\sigma^2(\theta)=\sigma(\theta\lambda)=\theta\lambda\lambda^{-1}=\theta$, it means $\sigma^2=id$.
And $\tau(\lambda)=\frac{1}{2}+\frac{1}{2}\theta^3\cdot\lambda^6=\frac{1}{2}+\frac{1}{2}i\sqrt{3}=\lambda$, so
$\tau^2(\theta)=\tau(\theta\lambda^2)=\theta\lambda^4$, $\tau^3(\theta)=\theta\lambda^6=\theta$, so $\tau^3=id$.
And, $\sigma\tau(\theta)=\sigma(\theta\lambda^2)=\theta\cdot\lambda^{-1}=\theta\cdot\lambda^2$, $(\sigma\tau)^2(\theta)=\sigma\tau(\theta\lambda^2)=\theta\lambda^{-3}=\theta$,
therefore $\sigma\tau\sigma\tau=id$, or $\sigma\tau\sigma=\tau^{-1}$. Now, $Gal(K/F)=\langle\sigma,\tau\rangle\cong D_6\cong S_3$.

Alternatively, if $Gal(K/\mathbb{Q})\cong\mathbb{Z}/6\mathbb{Z}$, the abelian group, then every subgroup is normal.
Therefore every intermidate field E satisfies E/F is Galois. (In other say, normal)
Since $\sqrt[6]{3}\cdot w^{11}\cdot\sqrt[6]{3}\cdot w=\sqrt[3]{3}\in K$, the field $\mathbb{Q}(\sqrt[3]{3})$ is the intermidate field,
but as we know, the minimal polynomial of $\sqrt[3]{3}$ is x^3-3 over $\mathbb{Q}$, but $\mathbb{Q}(\sqrt[3]{3})$ does not have other roots of x^3-3. So, it is contradiction.
By the knowledge in the group theory, there are only two group of order 6, S_3 and $\mathbb{Z}/6\mathbb{Z}$, so $Gal(K/\mathbb{Q})\cong S_3$. ∎

「Galois group of x^6+3 over $F = \mathbb{Q}$ or $\mathbb{F}_5$ or $\mathbb{F}_7$. (UCLA Qual, 2020 Fall)
For each case, determine the splitting field K and evaluate $[K:F]$.
And, find the Galois group $\mathrm{Gal}(K/F)$.
(Note: the field of characteristic 0 or finite field are perfect.).

2). $F = \mathbb{F}_5$

Suppose that $\alpha \in \overline{\mathbb{F}_5}$ satisfies $\alpha^2 = 3$. Then, $(\alpha^2)^3 + 3 = 30 \equiv 0 \mod 5$,
therefore x^2-3 divides x^6+3, and x^2-3 is irreducible over $\mathbb{F}_5$
because substituting $x = 0, 1, \cdots, 4$, $x^2-3 \neq 0$, by casework. ($\{x^2 \mid x \in \mathbb{F}_5\} = \{0,1,4\}$)
Let $K = \mathbb{F}_5(\alpha)$. Since x^2-3 is the minimal polynomial of α, $[K:\mathbb{F}_5] = 2$.
Given $a+b\alpha \in K$ $(a,b \in \mathbb{F}_5)$, observe that

$$(a+b\alpha)^6 = \underbrace{(a+b\alpha)^5}_{\text{Frobenius automorphism which fixes } \mathbb{F}_5} \cdot (a+b\alpha) = (a+b\alpha^5)\cdot(a+b\alpha) = (a+b\alpha)(a-b\alpha) = a^2 \underbrace{+2}_{=-3} b^2 = 2$$

if and only if:

4 18 22

$$\begin{cases} a = 0 \\ b = \pm 1 \end{cases} \quad \text{or} \quad \begin{cases} a = \pm 2 \\ b = \pm 2 \end{cases} \left(\Leftrightarrow \begin{cases} a = 2 \text{ or } 3 \\ b = 2 \text{ or } 3 \end{cases} \right)$$

So, the six distinct roots are contained in K, so K is splitting field of x^6+3,
and clearly it is separable, therefore $K/\mathbb{F}_5$ is Galois, with the Galois group

$$\mathrm{Gal}(K/\mathbb{F}_5) = \langle x \mapsto x^5 \rangle \cong \mathbb{Z}_2.$$

In actually, $K = \mathbb{F}_{5^2}$.

「Galois group of x^6+3 over $F=\mathbb{Q}$ or $\mathbb{F}_5$ or $\mathbb{F}_7$. (UCLA Qual, 2020 Fall)
For each case, determine the splitting field K and evaluate $[K:F]$.
And, find the Galois group $Gal(K/F)$.
(Note: the field of characteristic 0 or finite field are perfect.).

3). $F=\mathbb{F}_7$

「Galois group of x^3-2 over $\mathbb{F}_7$. (By Gu)

Since $0, 1, \cdots, 6$ are not root of x^3-2, so it is degree 3 Polynomial having no root in $\mathbb{F}_7$ therefore it is irreducible. ($\{x^3 \mid x \in \mathbb{F}_7\} = \{0, 1, 6\} \not\ni 2$)
Let $\alpha \in \mathbb{F}_7$ be a root of x^3-2. Then, $x^3-2 \in \mathbb{F}_7$ is the minimal polynomial of the α, by the irreducibility, so $K = \mathbb{F}_7(\alpha)$ satisfies that $[K : \mathbb{F}_7] = 3$.
Now, the extension K is $\mathbb{F}_7$-vector space generated by $\{1, \alpha, \alpha^2\}$,
Moreover, since $x^3 = 1 \iff x = 1, 2, 4 \in \mathbb{F}_7$,
therefore $\alpha, 2\alpha, 4\alpha \in K$ satisfies $\alpha^3 = 2$, $(2\alpha)^3 = 2$, $(4\alpha)^3 = 2$
So, K is the splitting field of x^3-2 over $\mathbb{F}_7$, Consequently,

$$\mathrm{Gal}(K/\mathbb{F}_7) = \langle a \mapsto a^7 \rangle \cong \mathbb{Z}/3\mathbb{Z}.$$

「Debris.

let $a + b\alpha + c\alpha^2 \in K$ $(a, b, c \in \mathbb{F}_7)$ be given. Then,

$$(a+b\alpha+c\alpha^2)^3 = (c\alpha^2)^3 + (a+b\alpha)^3 + 3(a+b\alpha)\cdot(c\alpha^2)(a+b\alpha+c\alpha^2)$$
$$= 4c^3 + (a^3 + 2b^3 + 3ab\alpha(a+b\alpha))$$
$$+$$

$x^3 = 1 \iff x = 1, 2, 4$
$2x^3 = 2$.
So, $x^3 = 2 \iff x = \sqrt[3]{2}\cdot\xi$.
$= \alpha \cdot 1$
α
2α
4α.

$$[a^2 + b^2\alpha^2 + c^2\alpha^4 + 2ab\alpha + 2ac\alpha^2 + 2bc\alpha^3]\cdot(a+b\alpha+c\alpha^2)$$
$$= [a^2 + 2bc + (2ab + 2c^2)\alpha + (b^2 + 2ac)\alpha^2]\cdot(a+b\alpha+c\alpha^2)$$
$$= a^3 + 2abc + (2a^2b + 2ac^2)\alpha + (ab^2 + 2a^2c)\alpha^2$$
$$+ a^2b\alpha + 2b^2c\alpha + (2ab^2 + 2bc^2)\alpha^2 + 2(b^3 + 2abc)$$
$$+ a^2c\alpha^2 + 2bc^2\alpha^2 + 4abc + 4c^3 + 2b^2c\alpha + 4ac^2\alpha$$
$$= a^3 + 2b^3 + 4c^3 + 3abc$$

[Galois group of $x^5+3x^3-2x^2-6$ over $\mathbb{Q}$.

Notice that

$$x^5+3x^3-2x^2-6 = x^2\cdot(x^3-2)+3(x^3-2) = (x^3-2)\cdot(x^2+3).$$

Since the irreducible factor x^3-2 has three roots $\sqrt[3]{2}$, $\sqrt[3]{2}\cdot e^{\frac{2\pi i}{3}}$, $\sqrt[3]{2}\cdot e^{\frac{4\pi i}{3}}$ and the irreducible factor x^2+3 has two roots $\pm\sqrt{-3}$,

The splitting field is

$$K = \mathbb{Q}\left(\sqrt[3]{2}, \sqrt[3]{2}\cdot e^{\frac{2\pi i}{3}}, \sqrt[3]{2}\cdot e^{\frac{4\pi i}{3}}, \sqrt{-3}, -\sqrt{-3}\right) = \mathbb{Q}(\sqrt[3]{2}, e^{\frac{2\pi i}{3}}, \sqrt{-3})$$

Moreover, since $\mathbb{Q}$ is perfect, so $K/\mathbb{Q}$ is Galois.

Meanwhile, notice that

$$e^{\frac{2\pi i}{3}} = -\frac{1}{2}+\frac{\sqrt{-3}}{2}$$

Therefore, $K = \mathbb{Q}(\sqrt[3]{2}, e^{\frac{2\pi i}{3}})$. Finally,

$$[\mathbb{Q}(\sqrt[3]{2}):\mathbb{Q}] = 3$$

and since $w = e^{\frac{2\pi i}{3}}$ satisfies w^2+w+1, so

$$[\mathbb{Q}(\sqrt[3]{2}, w):\mathbb{Q}(\sqrt[3]{2})] \leq 2$$

But since $w \in \mathbb{C}\setminus\mathbb{R}$ and $\mathbb{Q}(\sqrt[3]{2}) \subseteq \mathbb{R}$, so the degree is actually 2.

Since $Gal(K/\mathbb{Q}) \subseteq Sym(\{\sqrt[3]{2}, \sqrt[3]{2}w, \sqrt[3]{2}w^2\})$, the tower lemma gives $[\mathbb{Q}(\sqrt[3]{2}, w):\mathbb{Q}] = 6$, so $|Gal(K/\mathbb{Q})| = 6$, that is,

$$Gal(K/\mathbb{Q}) \cong S_3.$$

「Galois group of x^5+x-1 over $\mathbb{Q}$.

Notice that

$$x^5+x-1=(x^2+ax+1)(x^3+bx^2+cx-1)=(x^2-x+1)(x^3+x^2-1)$$

$$\begin{aligned}&= x^5+bx^4+cx^3-x^2\\ &\quad\; ax^4+abx^3+acx^2-ax\\ &\quad\quad\; x^3+bx^2+cx-1\end{aligned}$$

and both x^2-x+1 and x^3+x^2-1 are irreducible, because these do not have a root in $\mathbb{Q}$, and degree ≤ 3.

So, the splitting field K over $\mathbb{Q}$ satisfies

$$\mathrm{Gal}(K/\mathbb{Q}) \hookrightarrow S_2\times S_3 \cong C_2\times S_3.$$

Meanwhile, let K_1 and K_2 be the splitting fields of x^2-x+1 and x^3+x^2-1, respectively. Then, $K=K_1K_2$, and

$$K_1=\mathbb{Q}\left(\frac{1\pm\sqrt{-3}}{2}\right)=\mathbb{Q}(\sqrt{-3}),\ \text{so}\ \mathrm{Gal}(K_1/\mathbb{Q})\cong C_2.$$

And to investigate x^3+x^2-1, the discriminant is

$$D=-4\cdot(1)^3\cdot(-1)-27(-1)^2=-23$$

and $\sqrt{D}=\sqrt{-23}\notin\mathbb{Q}$, so $\mathrm{Gal}(K_2/\mathbb{Q})\cong S_3$

Meanwhile, the subgroup lattice of $\mathrm{Gal}(K/\mathbb{Q})$ is

$$\begin{tikzcd}[arrows=dash]
 & \mathrm{Gal}(K/\mathbb{Q}) \arrow[dl] \arrow[dr] & \\
 & & \mathrm{Gal}(\mathbb{Q}(\sqrt{-23})/\mathbb{Q}) \arrow[d] \\
\mathrm{Gal}(K_1/\mathbb{Q}) \arrow[dr] & & \mathrm{Gal}(K_2/\mathbb{Q}) \arrow[dl] \\
 & \mathrm{Gal}(\mathbb{Q}/\mathbb{Q}) &
\end{tikzcd}$$

$$\begin{tikzcd}[arrows=dash]
 & K=K_1K_2 \arrow[dl] \arrow[dr] & \\
 & & K_2 \arrow[d, "2"] \\
K_1 \arrow[dr] & & \mathbb{Q}(\sqrt{-23}) \arrow[dl] \\
 & \mathbb{Q} & \\
 & \rotatebox{90}{$=$} & \\
 & K^G &
\end{tikzcd}$$

Conti.

「Galois group of $x^p - x + a$ over $\mathbb{F}_p$ where $a \in \mathbb{F}_p \setminus \{0\}$.

First, we need to show that $x^p - x + a$ is irreducible and separable over $\mathbb{F}_p$.

To show the irreducibility, there are two way:

1). Let $\alpha \in \overline{\mathbb{F}_p}$ be a root of $x^p - x + a$. That is, $\alpha^p - \alpha + a = 0$.

Notice that $(\alpha+1)^p - (\alpha+1) + a = \alpha^p + 1 - (\alpha+1) + a = 0$, because on the field of characteristic p, $x \mapsto x^p$ is Homomorphism. Inductively, $\alpha, \alpha+1, \cdots, \alpha+p-1$ are roots of $x^p - x + a$. Moreover, these are mutually distinct, because $\alpha + i = \alpha + j \iff i = j$ but $\{1, \cdots, p-1\}$ are distinct in the field of characteristic p. Now, over $\overline{\mathbb{F}_p}$,

$$x^p - x + a = (x - \alpha)(x - (\alpha+1)) \cdots (x - (\alpha + p - 1))$$

If $x^p - x + a$ is reducible over $\mathbb{F}^p$, then $x^p - x + a = f(x) g(x)$ for some $f, g \in \mathbb{F}_p[x]$ and $\deg f, \deg g < p$. By the unique factorization over $\overline{\mathbb{F}_p}$,

$$f(x) = \prod_{i \in I} (x - (\alpha + i)) \text{ where } I \subsetneq \{0, 1, \cdots, p-1\}.$$

But, then $= x^{|I|} - \left(\sum_{i \in I} (\alpha + i)\right) \cdot x^{|I|-1} + \cdots,$

So $|I| \cdot \alpha + \sum_{i \in I} i \in \mathbb{F}_p$, but then $|I|^{-1} \cdot |I| \cdot \alpha = \alpha \in \mathbb{F}_p$. $(0 < |I| < p)$.

But, $\alpha \in \mathbb{F}_p \iff \alpha$ is a root of $x^p - x \Rightarrow a = 0$, it is contradiction.

Therefore, $x^p - x + a$ is irreducible, and separable because all roots are distinct.

Notice that $\mathbb{F}_p(\alpha)$ is the splitting field of $x^p - x + a$ over $\mathbb{F}$.

Now, $[\mathbb{F}_p(\alpha) : \mathbb{F}_p] = \deg(x^p - x + a) = p$, therefore

$$\mathrm{Gal}(\mathbb{F}_p(\alpha) / \mathbb{F}_p) = \langle x \mapsto x^p \rangle \cong C_p, \text{ the cyclic group of order } p$$

「Galois group of $\mathbb{Q}(\sqrt{2+\sqrt{2}})$ over $\mathbb{Q}$

First, since $x=\sqrt{2+\sqrt{2}} \Rightarrow x^4-4x^2+2=0$, the degree of minimal polynomial of $\sqrt{2+\sqrt{2}}$ divides 4. And, since $1\notin 2\mathbb{Z}$, $0,4,2\in 2\mathbb{Z}$ but $2\in 2^2\mathbb{Z}$, so the eisenstein criterion gives that the given polynomial is irreducible, therefore x^4-4x^2+2 is exactly the minimal polynomial of $\sqrt{2+\sqrt{2}}$.

By the fundamental theorem of Algebra, we can factor x^4-4x^2+2 as linear factors, as

$$x^4-4x^2+2=(x^2-(+2+\sqrt{2}))(x^2-(+2-\sqrt{2}))$$
$$=(x-\sqrt{2+\sqrt{2}})(x+\sqrt{2+\sqrt{2}})(x-\sqrt{2-\sqrt{2}})(x+\sqrt{2-\sqrt{2}})$$

To show that $\mathbb{Q}(\sqrt{2+\sqrt{2}})$ is splitting field of x^4-4x^2+2, enough to show that $\sqrt{2-\sqrt{2}}\in\mathbb{Q}(\sqrt{2+\sqrt{2}})$.

observe that: $\sqrt{2}=(\sqrt{2+\sqrt{2}})^2-2$ and

$$(\sqrt{2+\sqrt{2}})\cdot(\sqrt{2-\sqrt{2}})=\sqrt{2}, \text{ so } \sqrt{2-\sqrt{2}}=\frac{\sqrt{2}}{\sqrt{2+\sqrt{2}}}=\sqrt{2+\sqrt{2}}-\frac{2}{\sqrt{2+\sqrt{2}}}$$

therefore $\sqrt{2-\sqrt{2}}\in\mathbb{Q}(\sqrt{2+\sqrt{2}})$. And, the formal derivative $4x^3-8x=4x(x^2-2)$ has roots as $0, \pm\sqrt{2}$, so that x^4-4x^2+2 is separable.

This proves $\mathbb{Q}(\sqrt{2+\sqrt{2}})/\mathbb{Q}$ is Galois, and has degree of 4.

Now, consider an automorphism on $\mathbb{Q}(\sqrt{2+\sqrt{2}})$ as

$$\sigma: \underset{\alpha}{\underset{\parallel}{\sqrt{2+\sqrt{2}}}} \mapsto \underset{\beta}{\underset{\parallel}{\sqrt{2-\sqrt{2}}}}.$$

$\alpha^4-4\alpha^2+2=0$ / $\beta=$

$\alpha^3-4\alpha=-\frac{2}{\alpha}$

Then, $\beta=\alpha^3-3\alpha$, so $\qquad \beta^4-4\beta^2=-2 \qquad -\frac{2}{\beta}+\beta$

$$\sigma^2(\alpha)=\sigma(\beta)=\sigma(\alpha)^3-3\sigma(\alpha)=\beta^3-3\beta=\beta-\frac{2}{\beta}=-\alpha$$

So σ has order 4, therefore $\mathrm{Gal}((\mathbb{Q}(\sqrt{2+\sqrt{2}})/\mathbb{Q})=\langle\sigma\rangle\cong\mathbb{Z}/4\mathbb{Z}$.

$\mathbb{Q}(\sqrt{2+\sqrt{2}})$
$|$
$\mathbb{Q}(\sqrt{2})$ fixed by $\langle\sigma^2\rangle$
$|$
$\mathbb{Q}$

Note: $\cos 2\alpha=\cos^2\alpha-\sin^2\alpha=\cos^2\alpha+\cos^2\alpha-1 \Rightarrow \cos^2\alpha=\frac{1+\cos 2\alpha}{2}=1-\sin^2\alpha$.

So, $\cos\frac{\pi}{8}=\sqrt{\frac{1}{2}+\frac{1}{2}\cdot\frac{\sqrt{2}}{2}}=\frac{1}{2}\sqrt{2+\sqrt{2}}$ and $\sin\frac{\pi}{8}=\frac{1}{2}\sqrt{2-\sqrt{2}}$

Biquadratic criterion: $2^2-2=2$ is not square.

Galois group of $\mathbb{Q}(\sqrt{3+\sqrt{3}})$ over $\mathbb{Q}$

Galois group of $\mathbb{Q}(\sqrt{p}+\sqrt{q})$ over $\mathbb{Q}$, where p and q are distinct prime (UCLA, S학기, 2023)

First, $[\mathbb{Q}(\sqrt{p}) : \mathbb{Q}] = 2$ because x^2-p has $\sqrt{p}$ as root and it is irreducible, since x^2-p has no root in $\mathbb{Q}$.

And, $[\mathbb{Q}(\sqrt{p},\sqrt{q}) : \mathbb{Q}(\sqrt{p})] = 2$, because x^2-q has $\sqrt{q}$ as root and $\sqrt{q} \notin \mathbb{Q}(\sqrt{p})$, since: if $\sqrt{q} \in \mathbb{Q}(\sqrt{p})$, then $\sqrt{q} = a + b\sqrt{p}$, it implies

$$q = a^2 + b^2 p + 2ab\sqrt{p} \Rightarrow \sqrt{p} = \frac{a^2+b^2p}{2ab} \quad (a, b \neq 0)$$

Since $\sqrt{p} \notin \mathbb{Q}$, it is contradiction.

Now, to show $\mathbb{Q}(\sqrt{p}+\sqrt{q}) = \mathbb{Q}(\sqrt{p},\sqrt{q})$, observe that

$$\sqrt{p}+\sqrt{q} \in \mathbb{Q}(\sqrt{p},\sqrt{q}) \Rightarrow \mathbb{Q}(\sqrt{p}+\sqrt{q}) \subseteq \mathbb{Q}(\sqrt{p},\sqrt{q}).$$

By the tower lemma, $[\mathbb{Q}(\sqrt{p},\sqrt{q}) : \mathbb{Q}] = [\mathbb{Q}(\sqrt{p},\sqrt{q}) : \mathbb{Q}(\sqrt{p})][\mathbb{Q}(\sqrt{p}) : \mathbb{Q}] = 4$, enough to show that $[\mathbb{Q}(\sqrt{p}+\sqrt{q}) : \mathbb{Q}] = 4$.

If $[\mathbb{Q}(\sqrt{p}+\sqrt{q}) : \mathbb{Q}] = 1$, then $\sqrt{p}+\sqrt{q} \in \mathbb{Q}(\sqrt{p}+\sqrt{q}) = \mathbb{Q}$, it is contradiction.

If $[\mathbb{Q}(\sqrt{p}+\sqrt{q}) : \mathbb{Q}] = 3$, then

$$4 = [\mathbb{Q}(\sqrt{p},\sqrt{q}) : \mathbb{Q}(\sqrt{p}+\sqrt{q})][\mathbb{Q}(\sqrt{p}+\sqrt{q}) : \mathbb{Q}] = n \cdot 3 \quad \text{for some } n \in \mathbb{N},$$

but trivially $3 \nmid 4$, it contradicts

If $[\mathbb{Q}(\sqrt{p}+\sqrt{q}) : \mathbb{Q}] = 2$, then $\sqrt{p}+\sqrt{q}$ is a root of degree 2 polynomial over 2, that is, $(\sqrt{p}+\sqrt{q})^2 + a(\sqrt{p}+\sqrt{q}) + b = 0$ for some $a, b \in \mathbb{Q}$,

$$= \sqrt{pq} + \sqrt{p} + \sqrt{q} + p + q + b = 0$$ a는 어디에 팔아먹었나?

$$\Rightarrow pq = (\sqrt{p}+\sqrt{q})^2 + (p+q+b)^2 = (p+q+b)^2 + p + q + 2\sqrt{pq}$$

$$\Rightarrow \sqrt{pq} \in \mathbb{Q}$$

But it is contradiction, so, $\mathbb{Q}(\sqrt{p}+\sqrt{q}) = \mathbb{Q}(\sqrt{p},\sqrt{q})$.

Clearly, $\mathbb{Q}(\sqrt{p},\sqrt{q})$ is splitting field of $(x^2-p)(x^2-q)$, $\mathbb{Q}(\sqrt{p}+\sqrt{q})/\mathbb{Q}$ is Galois, and define

$$\sigma : \begin{cases} \sqrt{p} \mapsto -\sqrt{p} \\ \sqrt{q} \mapsto \sqrt{q} \end{cases} \quad \text{and} \quad \tau : \begin{cases} \sqrt{p} \mapsto \sqrt{p} \\ \sqrt{q} \mapsto -\sqrt{q}. \end{cases}$$

These are well-defined automorphisms of $\mathbb{Q}(\sqrt{p}+\sqrt{q})$ by the Galois theorem.

Moreover, $\mathrm{Gal}(\mathbb{Q}(\sqrt{p}+\sqrt{q})/\mathbb{Q}) = \langle \sigma, \tau \rangle \cong \mathbb{Z}_2 \times \mathbb{Z}_2$. ☒

[Galois group of $\mathbb{Q}(\sqrt{p}+\sqrt{q})$ over $\mathbb{Q}$, where p and q are distinct prime, Revised.

First, $[\mathbb{Q}(\sqrt{p}):\mathbb{Q}]=2$, because x^2-p is irreducible over $\mathbb{Q}$, since x^2-p has no root in $\mathbb{Q}$. And, $[\mathbb{Q}(\sqrt{p},\sqrt{q}):\mathbb{Q}(\sqrt{p})]=2$, because x^2-q has no root in $\mathbb{Q}(\sqrt{p})$. Specifically, if x^2-q has a root in $\mathbb{Q}(\sqrt{p})$, that is, $\sqrt{q}\in\mathbb{Q}(\sqrt{p})$, then $\sqrt{q}=a+b\sqrt{p}$ for some $a,b\in\mathbb{Q}$.

It implies
$$q=a^2+b^2p+2ab\sqrt{p},$$

but since $\sqrt{p}\in\mathbb{Q}$, either a or b must be zero.

But, $a=0$ means $\sqrt{q}=b\sqrt{p} \Rightarrow q=b^2p$ is not prime when $b\neq 1$, and $q=p$ is contradiction to q and p are distinct, so $b=1$ false.

And, $b=0$ means $\sqrt{q}=a\in\mathbb{Q}$, contradiction. Therefore, by the tower lemma,
$$[\mathbb{Q}(\sqrt{p},\sqrt{q}):\mathbb{Q}]=[\mathbb{Q}(\sqrt{p},\sqrt{q}):\mathbb{Q}(\sqrt{p})][\mathbb{Q}(\sqrt{p}):\mathbb{Q}]=4.$$

To show $\mathbb{Q}(\sqrt{p}+\sqrt{q})=\mathbb{Q}(\sqrt{p},\sqrt{q})$, observe that
$$\sqrt{p}+\sqrt{q}\in\mathbb{Q}(\sqrt{p},\sqrt{q}) \Rightarrow \mathbb{Q}(\sqrt{p}+\sqrt{q})\subseteq\mathbb{Q}(\sqrt{p},\sqrt{q})$$

And, observe that
$$p-q=\sqrt{p}^2-\sqrt{q}^2=(\sqrt{p}+\sqrt{q})(\sqrt{p}-\sqrt{q}) \Rightarrow \sqrt{p}-\sqrt{q}=\frac{p-q}{\sqrt{p}+\sqrt{q}}\in\mathbb{Q}(\sqrt{p}+\sqrt{q})$$

So, $\sqrt{p}=\frac{1}{2}(\sqrt{p}+\sqrt{q}+\sqrt{p}-\sqrt{q})$ and $\sqrt{q}=\frac{1}{2}(\sqrt{p}+\sqrt{q}-(\sqrt{p}-\sqrt{q}))$

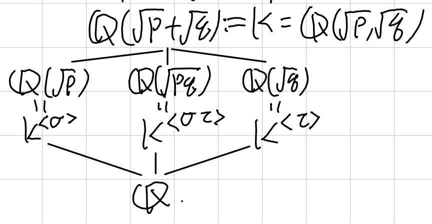
are contained in $\mathbb{Q}(\sqrt{p}+\sqrt{q})$. Thus $\mathbb{Q}(\sqrt{p}+\sqrt{q})=\mathbb{Q}(\sqrt{p},\sqrt{q})$.

And, $\mathbb{Q}(\sqrt{p},\sqrt{q})$ is the splitting field of $(x^2-p)(x^2-q)$ over $\mathbb{Q}$. Therefore, $\mathbb{Q}(\sqrt{p},\sqrt{q})/\mathbb{Q}$ is Galois, so it has four automorphism that fixes $\mathbb{Q}$.

Now, we can explicit two automorphisms:
$$\sigma:\begin{cases}\sqrt{p}\mapsto\sqrt{p}\\ \sqrt{q}\mapsto-\sqrt{q}\end{cases} \quad\text{and}\quad \tau:\begin{cases}\sqrt{p}\mapsto-\sqrt{p}\\ \sqrt{q}\mapsto\sqrt{q}\end{cases}$$

Then, $\sigma^2=\tau^2=(\sigma\tau)^2=id$ by just simple calculation.

Consequently, $Gal((\mathbb{Q}(\sqrt{p},\sqrt{q})/\mathbb{Q})=\langle\sigma,\tau\rangle\cong\mathbb{Z}_2\times\mathbb{Z}_2$,



「Galois group of $\mathbb{Q}(x)/\mathbb{Q}(x^2)$

「Galois group of $\mathbb{Q}(x,y)/\mathbb{Q}(xy)$